EduFin SQL Business Challenge

Phase 3 — Wednesday Submission

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STEP 3A — The Scale Shift (Comparison Analysis)

```
SELECT
    dc.city_name AS `City`,
    COUNT(*) AS `Total Loans`,
    SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS `Defaulted Loans`,
    CONCAT(ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) / COUNT(*) * 100, 2), '%') AS
`Default Rates (%)`
FROM loans_small l
INNER JOIN customers_small c ON l.customer_id = c.customer_id
INNER JOIN dim_city_small dc ON c.city_id = dc.city_id
GROUP BY dc.city_name
HAVING COUNT(*) >= 100
ORDER BY SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) / COUNT(*) DESC
LIMIT 10;

SELECT
    dc.city_name AS `City`,
    COUNT(*) AS `Total Loans`,
    SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS `Defaulted Loans`,
    CONCAT(ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) / COUNT(*) * 100, 2), '%') AS
`Default Rates (%)`
FROM loans_national l
INNER JOIN customers_national c ON l.customer_id = c.customer_id
INNER JOIN dim_city_national dc ON c.city_id = dc.city_id
GROUP BY dc.city_name
HAVING COUNT(*) >= 100
ORDER BY SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) / COUNT(*) DESC
LIMIT 10;
```

STEP 3B — CIBIL Score Distribution Among Defaulters

```
WITH cibil_data AS (
    SELECT
        c.customer_id,
        c.cibil_score,
        c.full_name,
        l.loan_status,
        CASE
            WHEN c.cibil_score >= 750 AND c.cibil_score <= 900 THEN 'Excellent (750-900)'
            WHEN c.cibil_score >= 650 THEN 'Good (650-749)'
            WHEN c.cibil_score >= 550 THEN 'Fair (550-649)'
            ELSE 'Poor (<550)'
        END AS `CIBIL Brackets`
    FROM customers c
    INNER JOIN loans l
        ON c.customer_id = l.customer_id
)
SELECT
    `CIBIL Brackets`,
    COUNT(*) AS `Loans`,
    SUM(CASE WHEN loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS `Defaulted Loans`,
    CONCAT(ROUND(SUM(CASE WHEN loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2), '%') AS
`Default Rate (%)`
FROM cibil_data
GROUP BY `CIBIL Brackets`
ORDER BY SUM(CASE WHEN loan_status = 'Defaulted' THEN 1 ELSE 0 END) DESC;
```

CIBIL Brackets	Loans	Defaulted Loans	Default Rate (%)
Good (650-749)	2364	250	10.58%
Fair (550-649)	1218	217	17.82%
Excellent (750-900)	1278	84	6.57%
Poor (<550)	140	39	27.86%

STEP 3C — Open Metric Design: "Risk Efficiency"

```
WITH cibil_data AS (
    SELECT
        l.loan_status,
        CAST(l.loan_amount AS DECIMAL) AS loan_amount,
        CASE
            WHEN c.cibil_score >= 750 AND c.cibil_score <= 900 THEN 'Excellent (750-900)'
            WHEN c.cibil_score >= 650 THEN 'Good (650-749)'
            WHEN c.cibil_score >= 550 THEN 'Fair (550-649)'
            ELSE 'Poor (<550)'
        END AS cibil_bracket
    FROM customers c
    INNER JOIN loans l
        ON c.customer_id = l.customer_id
),
cibil_metrics AS (
    SELECT
        cibil_bracket,
        SUM(CASE WHEN loan_status = 'Defaulted' THEN loan_amount ELSE 0 END) AS defaulted_amount,
        SUM(loan_amount) AS total_portfolio,
        ROUND(SUM(CASE WHEN loan_status = 'Defaulted' THEN loan_amount ELSE 0 END) * 100.0 / SUM(loan_amount),
2) AS loss_rate
    FROM cibil_data
    GROUP BY cibil_bracket
)
SELECT
    cibil_bracket AS `CIBIL Bracket`,
    CONCAT(ROUND(defaulted_amount / 10000000, 2), ' Cr') AS `Total Defaulted Amount`,
    CONCAT(ROUND(total_portfolio / 10000000, 2), ' Cr') AS `Total Portfolio Value`,
    CONCAT(loss_rate, '%') AS `Loss Rate (%)`
FROM cibil_metrics
ORDER BY loss_rate DESC;
```

STEP 3D — Income and Employment Risk Profile

```
-- Analyze loan portfolio and defaults by employment type
WITH employment_stats AS (
    SELECT
        c.employemnet_type AS employment_type,
        COUNT(*) AS total_loans,
        SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS default_count,
        ROUND(SUM(CAST(l.loan_amount AS DECIMAL)) / 10000000.0, 2) AS portfolio_amount_cr,
        ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN CAST(l.loan_amount AS DECIMAL) ELSE 0 END) /
10000000.0, 2) AS defaulted_amount_cr,
        ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS
default_rate
    FROM customers c
    INNER JOIN loans l
```

```
        ON c.customer_id = l.customer_id
    GROUP BY c.employeemnet_type
    HAVING COUNT(*) > 0
)
SELECT
    COALESCE(employment_type, 'Unknown') AS `Employment Type`,
    CONCAT(ROUND(portfolio_amount_cr, 2), ' Cr') AS `Portfolio Amount`,
    CONCAT(ROUND(defaulted_amount_cr, 2), ' Cr') AS `Defaulted Amount`,
    total_loans AS `Total Loans`,
    default_count AS `Default Count`,
    CONCAT(default_rate, '%') AS `Default Rate (%)`
FROM employment_stats
ORDER BY default_count DESC;

-- Analyze loan portfolio and defaults by income brackets
WITH income_brackets AS (
    SELECT
        customer_id,
        CASE
            WHEN CAST(annual_income AS DOUBLE) < 300000 THEN 'Low Income (<3L)'
            WHEN CAST(annual_income AS DOUBLE) >= 300000 AND CAST(annual_income AS DOUBLE) <= 800000 THEN 'Mid
(3L-7L)'
            ELSE 'High Income (>8L)'
        END AS income_bracket
    FROM customers
),
portfolio_stats AS (
    SELECT
        ib.income_bracket,
        SUM(l.loan_amount) / 10000000.0 AS portfolio_amount_cr,
        SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount * 1.0 ELSE 0 END) / 10000000.0 AS
defaulted_amount_cr,
        COUNT(*) AS total_loans,
        SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS default_count,
        ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) / COUNT(*) * 100, 2) AS default_rate
    FROM income_brackets ib
    INNER JOIN loans l
        ON ib.customer_id = l.customer_id
    GROUP BY ib.income_bracket
)
SELECT
    income_bracket AS `Income Bracket`,
    CONCAT(ROUND(portfolio_amount_cr, 2), ' Cr') AS `Portfolio Amount`,
    CONCAT(ROUND(defaulted_amount_cr, 2), ' Cr') AS `Defaulted Amount`,
    total_loans AS `Loans`,
    default_count AS `Default Count`,
    CONCAT(default_rate, '%') AS `Default Rate (%)`
FROM portfolio_stats
ORDER BY default_count DESC;
```

STEP 3E — Prioritized Collections Target List (Final Output)

```
WITH customer_defaults AS (
    SELECT
        c.customer_id,
        c.full_name,
        c.phone_number,
        c.email_address,
        c.cibil_score,
        COUNT(l.loan_id) AS total_loans,
        SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS default_count,
        ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN CAST(l.loan_amount AS DECIMAL) ELSE 0 END) /
100000.0, 2) AS default_amount,
        ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS
default_rate,
        CASE
            WHEN c.cibil_score >= 750 THEN 0.5
            WHEN c.cibil_score >= 700 THEN 0.6
            WHEN c.cibil_score >= 650 THEN 0.7
            WHEN c.cibil_score >= 550 THEN 0.85
            ELSE 1.0
        END AS cibil_risk_multiplier
    FROM customers c
    INNER JOIN loans l
        ON c.customer_id = l.customer_id
    GROUP BY
        c.customer_id,
        c.full_name,
        c.phone_number,
        c.email_address,
        c.cibil_score
),
scored AS (
    SELECT
        *,
        ROUND(
            (default_amount * default_rate / 100.0) +
            (default_count * 2.5) +
            (cibil_risk_multiplier * 5),
            2
        ) AS priority_score
    FROM customer_defaults
),
segmented AS (
    SELECT
        *,
        CASE
            WHEN priority_score >= 25 THEN 'Critical'
            WHEN priority_score >= 18 THEN 'High'
            WHEN priority_score >= 10 THEN 'Medium'
            ELSE 'Low'
        END AS risk_segment,
```

```
        CASE
            WHEN priority_score >= 25 THEN 'Escalate to Recovery Agency + Legal'
            WHEN priority_score >= 18 THEN 'Aggressive Collections Calls + Payment Plan'
            WHEN priority_score >= 10 THEN 'Standard Collections Calls'
            ELSE 'SMS + Email Reminder'
        END AS recommended_action
    FROM scored
)
SELECT
    customer_id AS `Customer ID`,
    full_name AS `Name`,
    phone_number AS `Contact`,
    email_address AS `Email`,
    CONCAT(default_amount, ' L') AS `Defaulted Amount`,
    default_count AS `Default Count`,
    CONCAT(default_rate, '%') AS `Default Rate (%)`,
    cibil_score AS `CIBIL Score`,
    priority_score AS `Priority Score`,
    risk_segment AS `Risk Segment`,
    recommended_action AS `Recommend Action`
FROM segmented
ORDER BY priority_score DESC
LIMIT 50;
```

Customer ID	Name	Contact	Email	Defaulted Amount	Default Count	Default Rate (%)	CIBIL Score	Priority Score	Risk Segment	Recommend Action
1047	Raghav Dubey	918494902096	raghav.dubey194@yahoo.com	15.07 L	3	100.00%	486	27.57	Critical	Escalate to Recovery Agency + Legal
4620	Pushti Doctor	918092197558	pushti.doctor209@outlook.com	10.28 L	3	100.00%	637	22.03	High	Aggressive Collections Calls + Payment Plan
4724	Ishani Joshi	917231771111	ishani.joshi420@outlook.com	9.68 L	3	100.00%	583	21.43	High	Aggressive Collections Calls + Payment Plan
2935	Mohammed Mammen	918820058128	mohammed.mammen602@outlook.com	11.40 L	2	100.00%	557	20.65	High	Aggressive Collections Calls + Payment Plan
3102	Balveer Swamy	918071485656	balveer.swamy341@gmail.com	11.06 L	2	100.00%	560	20.31	High	Aggressive Collections Calls + Payment Plan
2463	Hamsini Wadhwa	917596992261	hamsini.wadhwa608@outlook.com	9.93 L	2	100.00%	541	19.93	High	Aggressive Collections Calls + Payment Plan
4347	Max Pillay	919575772783	max.pillay122@hotmail.com	11.70 L	2	100.00%	730	19.70	High	Aggressive Collections Calls + Payment Plan
1666	Saumya Sarraf	918731863798	saumya.sarraf606@yahoo.com	11.51 L	2	100.00%	711	19.51	High	Aggressive Collections Calls + Payment Plan
1128	Ronith Anne	918875996993	ronith.anne920@outlook.com	10.18 L	2	100.00%	582	19.43	High	Aggressive Collections Calls + Payment Plan
960	Faqid Doshi	919099636955	faqid.doshi222@gmail.com	9.40 L	2	100.00%	534	19.40	High	Aggressive Collections Calls + Payment Plan
992	Jeet Randhawa	919140933057	jeet.randhawa655@yahoo.com	9.85 L	2	100.00%	580	19.10	High	Aggressive Collections Calls + Payment Plan
2998	Peter Sarma	919799532101	peter.sarma835@hotmail.com	10.37 L	2	100.00%	669	18.87	High	Aggressive Collections Calls + Payment Plan
3030	Yashvi Kuruvilla	919798989829	yashvi.kuruvilla313@outlook.com	10.07 L	2	100.00%	714	18.07	High	Aggressive Collections Calls + Payment Plan
4431	Amol Nayak	917476703358	amol.nayak830@yahoo.com	8.74 L	2	100.00%	630	17.99	Medium	Standard Collections Calls
3298	Yashasvi Sundaram	917611212814	yashasvi.sundaram746@outlook.com	12.97 L	2	66.67%	598	17.90	Medium	Standard Collections Calls
3728	Jai Tailor	917563087040	jai.tailor912@gmail.com	12.91 L	2	66.67%	582	17.86	Medium	Standard Collections Calls
3821	Dakshesh Purohit	919054366215	dakshesh.purohit663@hotmail.com	13.63 L	2	66.67%	685	17.59	Medium	Standard Collections Calls
3041	Bishakha Tella	919976412724	bishakha.tella605@hotmail.com	8.13 L	2	100.00%	627	17.38	Medium	Standard Collections Calls
2587	Yagnesh Luthra	917691857946	yagnesh.luthra998@yahoo.com	7.88 L	2	100.00%	644	17.13	Medium	Standard Collections Calls
29	Ganga Kala	917123786248	ganga.kala668@hotmail.com	7.68 L	2	100.00%	605	16.93	Medium	Standard Collections Calls

only showing top 20 rows

Custom Column

```
scored AS (
    SELECT
        *,
```

```
        ROUND(
            (default_amount * default_rate / 100.0) +
            (default_count * 2.5) +
            (cibil_risk_multiplier * 5),
            2
        ) AS priority_score
    FROM customer_defaults
),
segmented AS (
    SELECT
        *,
        CASE
            WHEN priority_score >= 25 THEN 'Critical'
            WHEN priority_score >= 18 THEN 'High'
            WHEN priority_score >= 10 THEN 'Medium'
            ELSE 'Low'
        END AS risk_segment,
        CASE
            WHEN priority_score >= 25 THEN 'Escalate to Recovery Agency + Legal'
            WHEN priority_score >= 18 THEN 'Aggressive Collections Calls + Payment Plan'
            WHEN priority_score >= 10 THEN 'Standard Collections Calls'
            ELSE 'SMS + Email Reminder'
        END AS recommended_action
    FROM scored
)

CASE
   WHEN priority_score >= 25 THEN 'Escalate to Recovery Agency + Legal'
   WHEN priority_score >= 18 THEN 'Aggressive Collections Calls + Payment Plan'
   WHEN priority_score >= 10 THEN 'Standard Collections Calls'
   ELSE 'SMS + Email Reminder'
END AS recommended_action
```

> CTEs are better because they break complex logic into readable, named stages rather than burying it in nested CASE statements. This makes debugging easier (you can inspect each CTE layer) and lets the database reuse intermediate results instead of recalculating them.

WHY Gate

1. If you added employment_type as a third factor in borrower_segment — would any customers move to a different segment? Pick one specific customer and show the re-classification logic.

Yes, but only if employment_type weight is high enough. Currently, risk_score = (default_amount × default_rate / 100) + (default_count × 2.5) + (CIBIL_multiplier × 5). Adding employment_type with weight 0.3 gives: composite_score = risk_score + (employment_weight × 0.3), where Government Employee = 0.5, Private = 0.75, Self-Employed = 1.0. *Example: Ishani Joshi (Currently High Risk, score 21.43)* - With employment_type: 21.43 + (0.75 × 0.3) = 21.65 → Still High Risk - With higher weight (1.0): 21.43 + (0.75 × 1.0) = 21.73 → Still High Risk *Why no movement: Employment_type adds only 0.15-0.30 points, but segment boundaries are 7 points apart (Critical ≥25, High 18-25, Medium 10-18, Low <10). For segment shifts, employment_type weight would need to be 1.0 or higher—making it equally important as CIBIL score.*

2. Your top 50 list: the collections team can make 200 calls per week. How do you prioritise the order within your top 50? Is priority_score alone sufficient?

I have ordered my top 50 list using a priority_score that combines three factors: financial impact (defaulted_amount × default_rate), default frequency (default_count × 2.5), and credit risk (CIBIL multiplier × 5). This multi-factor approach ensures the collections team—with 200 calls/week capacity—focuses on customers who are both high-loss AND hard-to-recover: Raghav Dubey ranks first with ₹15.07L defaulted and CIBIL 486, representing maximum recovery challenge. The ranking eliminates manual prioritization and allocates finite capacity to highest-impact cases.

3. Write the "Riskiest Borrower Profile" paragraph. Describe the type of customer who appears most frequently in your top segment.

The riskiest borrower profile is typically a self-employed or student customer with a low to mid CIBIL score and multiple past defaults. These borrowers often have higher defaulted loan amounts and repeated default behavior, indicating weak repayment stability. They appear most frequently in the top risk segment due to both financial exposure and unstable income sources.